

Nonlinear recovery relative to Kolmogorov width is equivalent to one-sided discretization

Candidate strong partial result for arXiv:2402.00848

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Status: strong partial result, likely valid. For every finite p , dimension N , and sample budget m , the worst-case constant in nonlinear recovery relative to the N th L_∞ Kolmogorov width is within explicit affine constants of the worst-case one-sided L_p discretization constant. Thus Open Problems 3.1 and 5.1 of the source are quantitatively equivalent. The common optimal growth rate and the linear recovery problem remain open.

1 Source questions and definitions

The source is I. Limonova, Yu. Malykhin, and V. Temlyakov, *One-sided discretization inequalities and sampling recovery*, arXiv:2402.00848. Let Ω be compact, let μ be a probability measure, and let $X \subset \mathcal{C}(\Omega)$ be N -dimensional. For sampling points $\xi = (\xi^1, \dots, \xi^m)$, put

$$D(X, \xi) := \sup_{0 \neq x \in X} \frac{\|x\|_{L_p(\mu)}}{\max_j |x(\xi^j)|},$$

with value $+\infty$ when the sampling map is not injective, and put $D_m(X) = \inf_\xi D(X, \xi)$. The universal constant is

$$D_{m,p}(N) := \sup_{(\Omega, \mu), \dim X = N} D_m(X).$$

For a compact class $\mathcal{F} \subset \mathcal{C}(\Omega)$, let $d_N(\mathcal{F}, L_\infty)$ be its N th Kolmogorov width. The nonlinear sampling error $\varrho_m^*(\mathcal{F}, L_p)$ is the infimum, over m sampling points and all reconstruction maps whose range lies in an at-most- m -dimensional linear subspace, of the worst L_p error. Define

$$R_{m,p}^*(N) := \sup_{(\Omega, \mu), \mathcal{F}} \frac{\varrho_m^*(\mathcal{F}, L_p)}{d_N(\mathcal{F}, L_\infty)},$$

where the supremum ranges over compact classes with nonzero width.

Open Problem 3.1 asks for the growth of m making $D_{m,p}(N)$ bounded by a constant depending only on p , for $2 < p < \infty$. Open Problem 5.1 asks for the growth making $R_{m,p}^*(N)$ bounded. Open Problem 5.2 asks the analogous question for linear reconstruction.

we discovered that the case $p = 2$, i.e. the $\text{LDI}(2,2)$

$$\|f\|_2 \leq C_2 \left(\sum_{j=1}^m \frac{1}{m} |f(\xi^j)|^2 \right)^{1/2}, \quad \forall f \in X_N, \quad m \leq C_1 N, \quad (3.2)$$

was obtained independently by a different method in the paper [1, Theorem 6.2], see also [33, Lemma 11].

Open problem 3.1. Under what condition on m does each N -dimensional subspace $X_N \subset \mathcal{C}(\Omega)$ have the property $X_N \in \mathcal{LD}(m, p, \infty, D)$, where $2 < p < \infty$ and $D = D(p)$:

$$\|f\|_p \leq D \max_{1 \leq j \leq m} |f(\xi^j)|, \quad \forall f \in X_N,$$

with an appropriate set of points $\{\xi^j\}_{j=1}^m$?

Comment 3.1. Theorem 1.3 from [23] on discretization of the uniform norm imply that in Open problem 3.1 we can always take $m = 9^N$, which means an exponential growth in N . Theorem 1.2 from [23] shows that for the subspace $\mathcal{T}(\Lambda_N)$ (see Corollary 2.1) we need $(N/e)e^{CN/D^2}$ points for the $\text{LDI}(\infty)$ to hold. Proposition 3.2 for $p = 2$ guarantees that we can take m of order N in the case $p \leq 2$. From Proposition 3.2 it follows that we can take $m \leq CN^{p/2} \log N$ in the case $p > 2$ and $m \leq CN^{p/2}$ for even p .

The following theorem from [33] is the $\text{LDI}(p,2)$.

Theorem 3.1 ([33, Theorem 13]). *Let $2 \leq p \leq \infty$. There exist positive constants $C_1 = 4$ and $C_2 = 83$ such that each N -dimensional subspace $X_N \subset$*

Figure 1: Open Problem 3.1 and the source's known bounds, official arXiv PDF, page 14.

in the case of Theorem 5.2 condition (5.1) are satisfied for any N -dimensional subspace of $\mathcal{C}(\Omega)$. In the case $p = 2$ the following inequality is known (see [50]): There exist two positive absolute constants b and B such that for any compact subset Ω of $\mathbb{R}^d$, any probability measure μ on it, and any compact subset $\mathbf{F}$ of $\mathcal{C}(\Omega)$ we have

$$\varrho_{bn}(\mathbf{F}, L_2(\Omega, \mu)) \leq B d_n(\mathbf{F}, L_\infty). \quad (5.7)$$

It is known (see [22] D.20. A lower bound]) and it follows from Corollary 2.1 that for discretization of the L_p -norm, $2 < p < \infty$, of functions in N -dimensional subspace we need at least $m > CN^{p/2}$ points. Thus, we can expect that the way of using an argument similar to Theorem 5.A will not allow us to obtain better than m of order $N^{p/2}$ in the inequality (5.6) for $2 < p < \infty$. Condition (5.1) is weaker than the combination of conditions A.1 and A.2. Therefore, there is a hope that we can get a better than m of order $N^{p/2}$ in the inequality (5.6) by using Theorem 5.2 if we solve **Open problem 3.1**.

Open problem 5.1. Let $2 < p < \infty$. What is the minimal growth of $m(N)$ on N , which guarantees the property: For any compact subset $\mathbf{F} \subset \mathcal{C}(\Omega)$ and for any probability measure μ on Ω

$$\varrho_{m(N)}^*(\mathbf{F}, L_p(\Omega, \mu)) \leq C d_N(\mathbf{F}, L_\infty), \quad 2 < p < \infty.$$

Open problem 5.2. Let $2 < p < \infty$. What is the minimal growth of $m(N)$ on N , which guarantees the property: For any compact subset $\mathbf{F} \subset \mathcal{C}(\Omega)$ and for any probability measure μ on Ω

$$\varrho_{m(N)}(\mathbf{F}, L_p(\Omega, \mu)) \leq C d_N(\mathbf{F}, L_\infty), \quad 2 < p < \infty.$$

Figure 2: Open Problems 5.1 and 5.2, official arXiv PDF, page 22.

2 Proof Intuition

The easy direction is the source's recovery argument: a sample set which norms every vector of an approximating subspace lets one convert small sampled residual into small L_p residual.

For the reverse direction, start with a subspace that no m samples norm well. Its L_p unit ball by itself has zero N th width, so it cannot test a width inequality. Thicken that ball by a compact family of uniformly small functions. The family is chosen canonically: pointwise clip each unit-ball element to the disk of radius ε . Given any proposed sample set, choose a unit vector whose sampled values are at most ε and subtract its clip. This erases every observation but leaves L_p norm at least $1 - \varepsilon$. The opposite function has the same zero data, so no algorithm can approximate both. Meanwhile the entire thickened class stays within ε in L_∞ of the original N -space.

3 Quantitative equivalence

For $\varepsilon > 0$, let $c_\varepsilon : \mathbb{C} \rightarrow \mathbb{C}$ be radial clipping onto the closed disk of radius ε :

$$c_\varepsilon(z) = \begin{cases} z, & |z| \leq \varepsilon, \\ \varepsilon z/|z|, & |z| > \varepsilon. \end{cases}$$

For real scalars this is ordinary clipping to $[-\varepsilon, \varepsilon]$. It acts pointwise on continuous functions.

Theorem 1. *For every $1 \leq p < \infty$ and all positive integers m, N ,*

$$D_{m,p}(N) - 1 \leq R_{m,p}^*(N) \leq 2D_{m,p}(N) + 1.$$

In particular, a function $m = m(N)$ gives a dimension-free one-sided discretization constant if and only if it gives a dimension-free nonlinear recovery-to-width constant.

Proof. We first prove the upper bound, including the short source mechanism. Fix a compact class $\mathcal{F}$ and $\eta > 0$, and choose an N -dimensional space X such that

$$\sup_{f \in \mathcal{F}} d(f, X)_\infty \leq (1 + \eta) d_N(\mathcal{F}, L_\infty).$$

Choose m samples with $D(X, \xi) \leq D_{m,p}(N) + \eta$. For each f , let $u \in X$ minimize the sampled ℓ_∞ residual, and choose $h \in X$ arbitrarily close to a best uniform approximant. Minimality gives

$$\max_j |(h - u)(\xi^j)| \leq 2\|f - h\|_\infty.$$

Consequently,

$$\|f - u\|_p \leq \|f - h\|_p + \|h - u\|_p \leq (1 + 2D(X, \xi))\|f - h\|_\infty.$$

Letting the approximation errors and η tend to zero yields $R_{m,p}^*(N) \leq 2D_{m,p}(N) + 1$.

For the converse, fix X and write $D = D_m(X)$. The claim is trivial when $D \leq 1$, so assume $1 < D < \infty$ and put $\varepsilon = D^{-1}$. Let

$$U = \{x \in X : \|x\|_p \leq 1\}, \quad H_\varepsilon = \{-c_\varepsilon \circ x : x \in U\}, \quad \mathcal{F}_X = U + H_\varepsilon.$$

The set U is compact in the uniform norm because X is finite-dimensional. Pointwise clipping is continuous in that norm, so H_ε and $\mathcal{F}_X$ are compact. Moreover, every $x + h \in \mathcal{F}_X$ is within $\|h\|_\infty \leq \varepsilon$ of X , whence

$$d_N(\mathcal{F}_X, L_\infty) \leq \varepsilon.$$

Now fix *arbitrary* m sampling points ξ . Since $D(X, \xi) \geq D$, compactness of the L_p unit sphere gives an $x \in X$ with

$$\|x\|_p = 1, \quad \max_j |x(\xi^j)| \leq D^{-1} = \varepsilon.$$

(If the sampling map has a kernel, take a unit vector in it.) Hence

$$f = x - c_\varepsilon \circ x \in \mathcal{F}_X, \quad f(\xi^j) = 0 \quad (1 \leq j \leq m),$$

and, because μ is a probability measure,

$$\|f\|_p \geq 1 - \|c_\varepsilon \circ x\|_p \geq 1 - \varepsilon.$$

The clipping map is odd, so $-f \in \mathcal{F}_X$ as well. Both functions produce the same zero sample vector. If an arbitrary reconstruction map returns g on that vector, the triangle inequality gives

$$\max\{\|f - g\|_p, \|-f - g\|_p\} \geq \|f\|_p \geq 1 - \varepsilon.$$

This holds for every sample set and every reconstruction map, even without a range restriction. Therefore

$$\frac{\varrho_m^*(\mathcal{F}_X, L_p)}{d_N(\mathcal{F}_X, L_\infty)} \geq \frac{1 - \varepsilon}{\varepsilon} = D - 1.$$

Taking the supremum over X proves the lower bound. If $D_m(X) = \infty$, the same construction with arbitrary $\varepsilon > 0$ gives an unbounded ratio, so the extended-real statement also follows. $\square$

4 Consequences, verification, and limitations

The theorem turns the source's sufficient implication into an if-and-only-if statement with constants:

$$D_{m,p}(N) \leq R_{m,p}^*(N) + 1, \quad R_{m,p}^*(N) \leq 2D_{m,p}(N) + 1.$$

Thus Open Problem 5.1 cannot have a better universal sampling growth than Open Problem 3.1. Any lower bound, logarithmic obstruction, or optimal theorem for one problem transfers immediately to the other.

The obstruction applies to every recovery rule, so if $R_{m,p}(N)$ denotes the linear-recovery analogue, then

$$D_{m,p}(N) - 1 \leq R_{m,p}^*(N) \leq R_{m,p}(N).$$

The missing reverse bound for $R_{m,p}(N)$ is substantive: the sampled ℓ_∞ best-approximation map used above is nonlinear, while a bounded linear inverse/extension can incur a dimension-dependent projection constant. Hence the proof does not solve Open Problem 5.2.

Verification. The script `code/verify_finite_obstruction.py` exhausts all 120 two-point sample sets for twenty rotations of a sixteen-point circular model. It computes the sampling-polytope vertices, constructs the pointwise cancellation, and checks the zero data, uniform perturbation bound, L_p lower bound, and ratio inequality. It ends with ALL FINITE OBSTRUCTION CHECKS PASSED. This is a sanity check; the proof above is exact and non-computational.

Upgrade attempts. Eight focused routes were pursued: even-moment tensorization, Lewis-position sparsification, rare-spike random sampling, VC hitting sets, symmetric lower examples, sparse-spike constructions, linear extension, and later-literature search. They did not close the known $N^{p/2}$ versus $N^{p/2} \log N$ gap for non-even p or produce a linear-recovery upper bound. Details are in the associated attempt note.

Novelty audit. The source PDF and TeX were inspected. Bounded arXiv searches through 12 August 2026 used the exact title, authors, open-problem language, “one-sided discretization”, “sampling recovery”, and the $N^{p/2}$ scale. They found the source, standard norm-discretization papers, and the 2026 Dai–Temlyakov survey, but no converse via a compact thickened unit ball and no explicit quantitative equivalence of the two worst-case constants. Novelty confidence is moderate and requires expert literature review.

Human-review recommendation. Check the compactness of the clipped family, the quantifier order (the class is fixed before sampling, while its indistinguishable pair may depend on the samples), and conventions for d_N and the range dimension in ϱ_m^* . These are the only delicate points; the norm estimates are elementary.

References

- [1] I. Limonova, Yu. Malykhin, and V. Temlyakov, *One-sided discretization inequalities and sampling recovery*, arXiv:2402.00848.
- [2] F. Dai and V. Temlyakov, *A survey on sampling recovery*, arXiv:2601.08787.